



NDMO Data Readiness Assessment

A structured self-assessment for Gulf financial institutions preparing for AI at scale. Evaluate your data foundation across six critical dimensions and receive a prioritised remediation roadmap.

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Contact: rodney@theenterpriseai.co.uk

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1. Why Data Readiness Before AI

85% of Saudi banks have an AI strategy. Fewer than 15% have the data foundation to execute it. The gap is not technology -- it is data classification, quality, and governance. NDMO's National Data Governance Interim Regulations and SDAIA's AI Ethics Principles both require demonstrable data governance before AI models can be deployed in regulated environments.

The Regulatory Landscape

Saudi Arabia's data governance requirements span multiple regulators: NDMO sets national standards for data classification and sharing; SDAIA governs AI ethics and data use; SAMA enforces financial sector data controls; and the PDPL (enacted September 2023) mandates personal data protection with significant penalties for non-compliance.

The AI Connection

Every AI model inherits the quality, bias, and governance posture of its training data. Without classification, you cannot control what data enters which model. Without lineage, you cannot audit model outputs. Without quality controls, your AI delivers confident wrong answers at scale.

2. The Six Dimensions of Data Readiness

1. Data Classification

Structured taxonomy aligned to NDMO's four-tier classification (Public, Restricted, Confidential, Top Secret). Every dataset tagged, every access control mapped, every cross-border transfer rule enforced. Without this, AI models cannot be risk-tiered.

2. Data Quality

Measured across six axes: completeness, accuracy, consistency, timeliness, validity, and uniqueness. Profiling at source, quality rules in pipelines, and automated monitoring dashboards. AI accuracy is bounded by input data quality.

3. Data Governance Operating Model

Roles, stewards, councils, and escalation paths. A Chief Data Officer mandate, domain data owners, and stewardship networks. Decision rights for data creation, modification, sharing, and deletion -- formalised and enforced.

4. NDMO & PDPL Compliance

Conformance with NDMO's National Data Governance Interim Regulations, Open Data Policy, and Data Sharing Regulations. PDPL readiness including consent management, data subject rights, cross-border transfer controls, and breach notification procedures.

5. Data Lineage & Traceability

End-to-end visibility from source systems through transformation to AI model inputs and outputs. Critical for model validation, regulatory audit trails, and explainability requirements under SAMA's expectations and the EU AI Act (for cross-border institutions).

6. Data Catalogue & Metadata

A searchable, governed inventory of all data assets -- business glossary, technical metadata, ownership, sensitivity labels, and usage policies. The foundation that makes all other dimensions discoverable and enforceable.

3. Self-Assessment Scorecard

Rate your institution across each dimension. Be candid -- this assessment is most valuable when it reflects reality, not aspiration.

Score	Maturity Level	Description
1	Ad Hoc	No formal process. Individuals manage data informally. No documentation.
2	Developing	Some processes exist but are inconsistent. Limited documentation. Reactive approach.
3	Defined	Formal policies and processes documented. Roles assigned. Partial implementation.
4	Managed	Processes measured and controlled. Regular reviews. Automation in place. Proactive monitoring.
5	Optimised	Continuous improvement. Industry-leading practices. Full automation. Real-time governance.

4. Assessment Matrix

For each dimension, assess your current state against the key indicators below. A score of 3 or above across all dimensions is the minimum threshold for enterprise AI deployment.

Dimension	Key Indicators	NDMO / Regulatory Link	AI Prerequisite
Classification	Taxonomy defined All datasets tagged Access controls enforced Cross-border rules applied	NDMO Data Classification Policy PDPL Art. 29	Model risk tiering Data access for training
Quality	Profiling complete Quality rules automated Monitoring dashboards live Remediation SLAs defined	SAMA Operational Risk NDMO Data Quality Standards	Model accuracy Bias detection Output reliability
Governance	CDO appointed Stewards active Councils meeting Policies enforced Escalation paths clear	NDMO National Data Governance Regulations	AI governance integration Decision rights for AI data use
NDMO / PDPL	Consent management Data subject rights Breach procedures Transfer controls DPO appointed	PDPL NDMO Open Data Data Sharing Regulations	Lawful AI training data Customer data in models

Lineage	Source-to-report tracing Transformation documented Impact analysis capability Audit trail complete	SAMA Model Risk NDMO Data Lifecycle	Model explainability Regulatory audit Bias tracing
Catalogue	Business glossary Technical metadata Ownership mapped Sensitivity labels Search enabled	NDMO Metadata Standards Data Inventory	Feature store readiness Data discovery for AI teams

5. Remediation Priorities by AI Maturity Stage

Stage 1: Pre-AI (Exploring)

Priority: Classification, Governance, Quality. Get the foundations in place before any AI investment. Appoint a CDO, classify your top 50 datasets, establish quality baselines. Time: 3-6 months.

Stage 2: Pilot AI (1-3 Use Cases)

Priority: Quality, Lineage, NDMO/PDPL. You are already building models -- ensure the data feeding them is profiled, traceable, and compliant. Automate quality rules in the pipeline. Time: 2-4 months.

Stage 3: Scaling AI (Enterprise Rollout)

Priority: Catalogue, Lineage, Governance maturity. At scale, data discovery and self-service become critical. Build the catalogue, mature the governance operating model, and ensure lineage covers all production models. Time: 4-8 months.

Stage 4: AI-Native Operations

Priority: Continuous improvement across all six dimensions. Real-time quality monitoring, automated classification, dynamic lineage, and governance-as-code. This is where data governance becomes a competitive advantage.

6. NDMO Regulatory Mapping

Key NDMO regulations and their data readiness implications for AI-deploying institutions.

NDMO Regulation	Core Requirements	AI Impact
National Data Governance Interim Regulations	Data governance framework CDO appointment Data management standards Annual compliance reporting	Mandatory foundation for any AI programme -- no governance, no AI deployment
Data Classification Policy	Four-tier classification Labelling standards Handling procedures Access controls per tier	Determines which data can be used for AI training, which models need additional controls

Data Sharing Regulations	Sharing agreements Purpose limitation Security requirements Cross-entity sharing controls	Critical for federated AI, multi-entity models, and third-party AI platform data flows
Open Data Policy	Public dataset publication Format standards API requirements Update schedules	Enrichment data for AI models. Obligation: ensure AI outputs using open data are attributed
Personal Data Protection Law (PDPL)	Consent Data subject rights Cross-border transfer Breach notification DPO appointment	Governs all AI models using customer data -- consent for profiling, right to explanation, automated decision safeguards

7. Next Steps

Self-Serve

Use this assessment to score your institution across all six dimensions. Identify your lowest-scoring areas and map them to the remediation priorities for your AI maturity stage. This gives you a board-ready view of data readiness gaps.

Guided Assessment

Enterprise.AI delivers a 4-week structured data readiness assessment that includes stakeholder interviews, automated data profiling, NDMO gap analysis, and a prioritised remediation roadmap with effort estimates and quick wins.

Next Steps

This guide is designed to be a practical, hands-on resource for implementation. Use the templates and checklists as starting points for your organization's specific needs.

We're here to help.

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Email: rodney@theenterpriseai.co.uk

For consultation on implementation, governance, and strategic deployment of AI initiatives.

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